Design and Analysis of a Three-Operational-Amplifier Instrumentation Amplifier (INA)

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Sensors and Applications Laboratory
Week 1 Report

1. Introduction

Instrumentation amplifiers (INAs) are highly specialized analog components engineered to amplify minute differential voltages while simultaneously providing excellent common-mode rejection and exceptionally high input impedance. Because of these robust performance metrics, the standard three-op-amp INA configuration is a staple in precision measurement systems, sensor signal conditioning, and biomedical applications.

2. Experiment

Designing and analysing an Instrumentation Amplifier (INA).

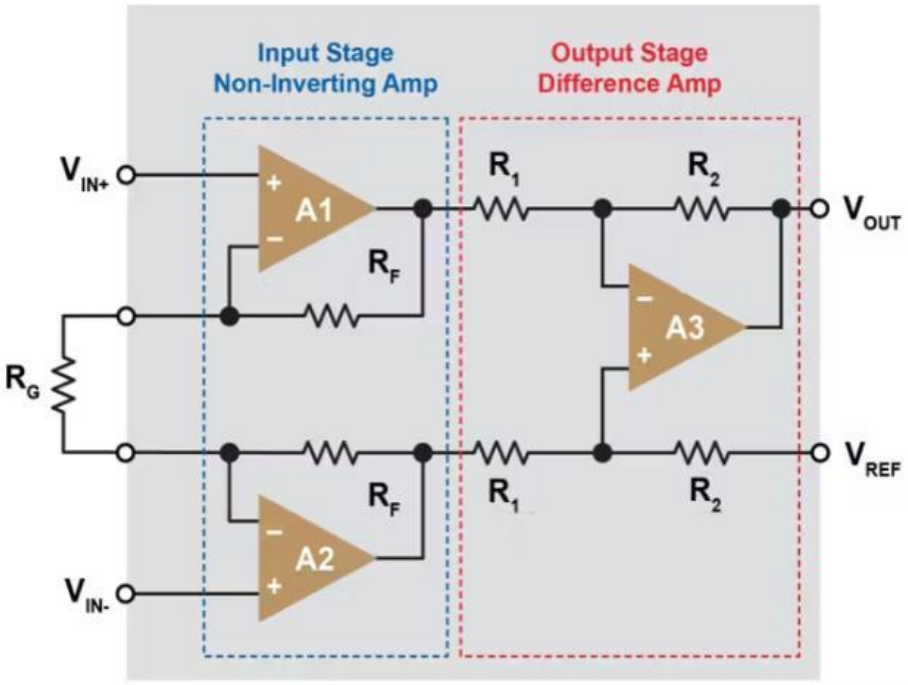


Figure 1: Three-op-amp instrumentation amplifier topology — input non-inverting stage (A1, A2) and output difference-amplifier stage (A3).

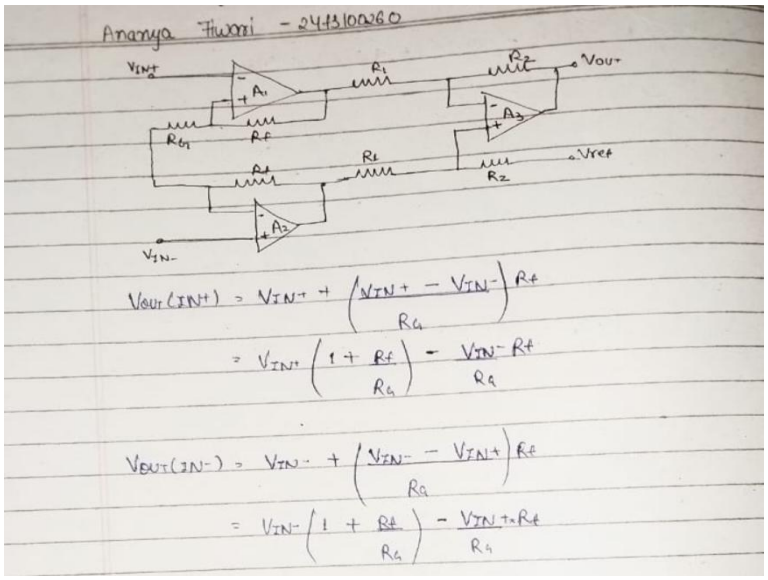
3. List of Components

- Op-Amp: MCP6004
- Breadboard
- Resistors (RF = R1 = R2 = R)
- Power Supply (ADALM1000)

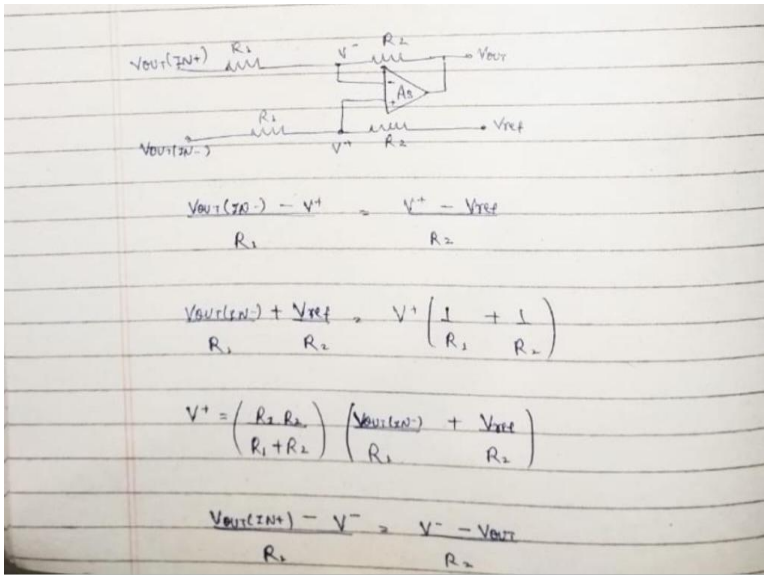
4. Design and Procedure

4.1 Gain Derivation ($R_F = R_1 = R_2 = R, V_{REF} = 0$)

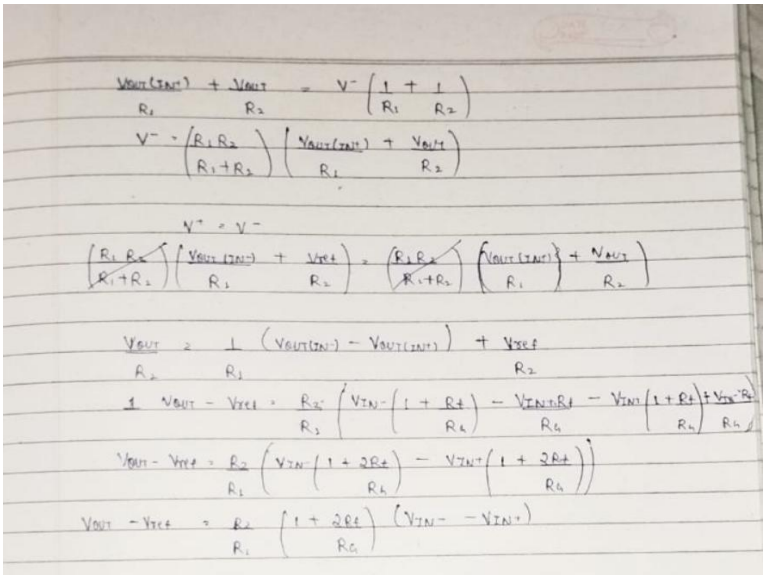
Task: derive an expression for the differential gain Ad by setting $R_F = R_1 = R_2 = R$ and $V_{REF} = 0$.



Handwritten derivation — input-stage output equations for $V_{OUT(IN+)}$ and $V_{OUT(IN-)}$.



Handwritten derivation — output difference-amplifier stage, solving for V+ and V−.



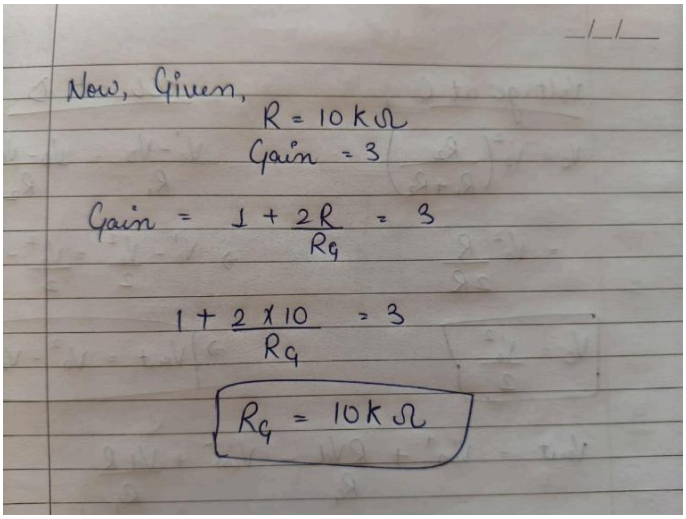
Handwritten derivation — combining the two stages to obtain the closed-form gain expression.

The derivation yields the standard three-op-amp INA transfer function:

$$V_{out} - V_{ref} = \frac{R_2}{R_1}\left(1 + \frac{2R}{R_G}\right)(V_{IN-} - V_{IN+})$$

4.2 Setting the Gain Resistor RG

With R = 10 kΩ, RG is calculated for a required differential gain of 3.



Hand calculation of R_G for $A_d = 3$, giving $R_G = 10\ k\Omega$.

4.3 Circuit Simulation (LTspice)

The designed circuit was simulated in LTspice to validate the analytical gain expression before hardware assembly.

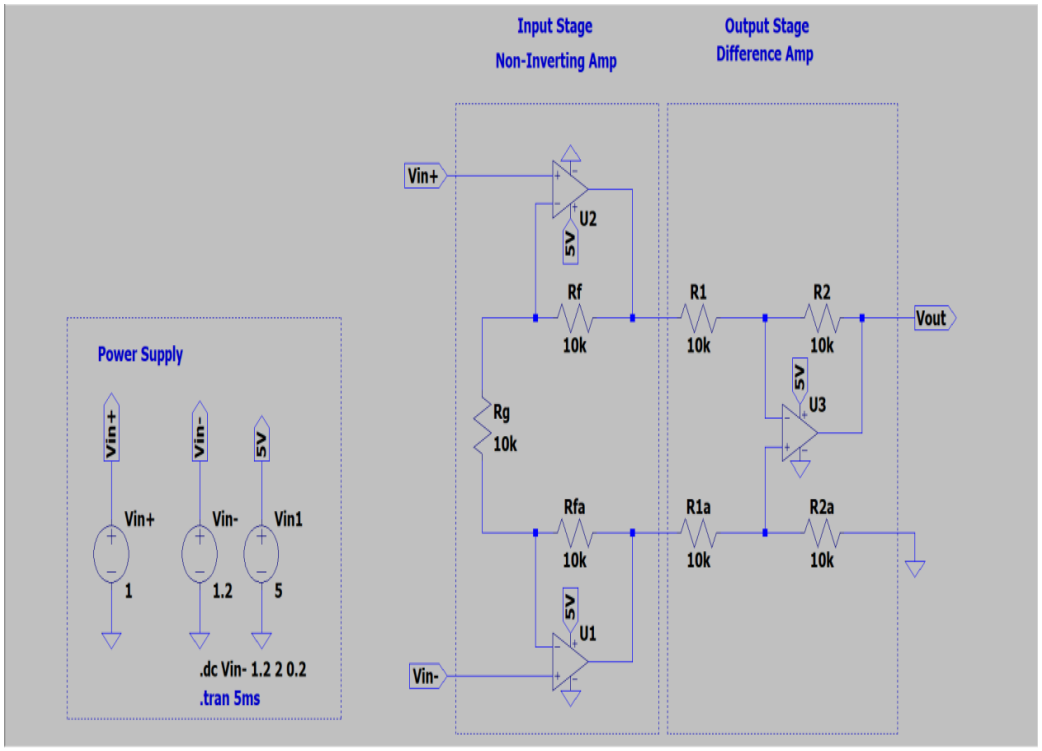


Figure 2: LTspice schematic of the three-op-amp INA ($R_F = R_1 = R_2 = R_G = 10\ k\Omega$).

4.4 DC Transfer Characteristic (Simulation)

The differential input voltage $V_d = V_{IN-} - V_{IN+}$ was swept and the corresponding V_{OUT} recorded to extract the differential gain Ad from the simulation.

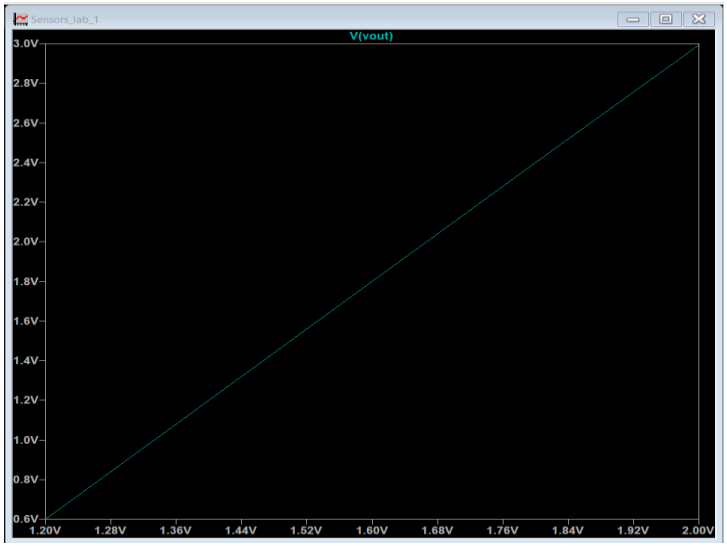


Figure 3: Simulated DC transfer characteristic, V(vout) vs. Vin−, confirming a linear response of slope ≈ 3.

5. Hardware Assembly

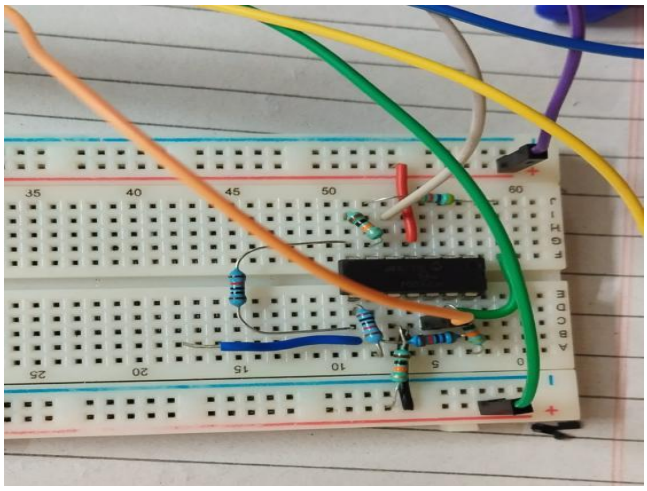


Figure 4: Breadboard hardware implementation of the INA using the MCP6004 quad op-amp.

Table 1: DC Transfer Characteristic Results

VIN+ (V)	VIN− (V)	Vd (V)	Vout,sim (V)	Vout,exp (V)
1.0	1.2	0.2	0.60	0.59
1.0	1.4	0.4	1.20	1.15
1.0	1.6	0.6	1.80	1.75
1.0	1.8	0.8	2.40	1.35
1.0	2.0	1.0	3.00	2.96

6. Common-Mode Response ($V_{IN+} = V_{IN-} = 1.5\,V$)

With both inputs tied to the same 1.5 V common-mode level, the residual output was measured to characterise common-mode gain and CMRR.

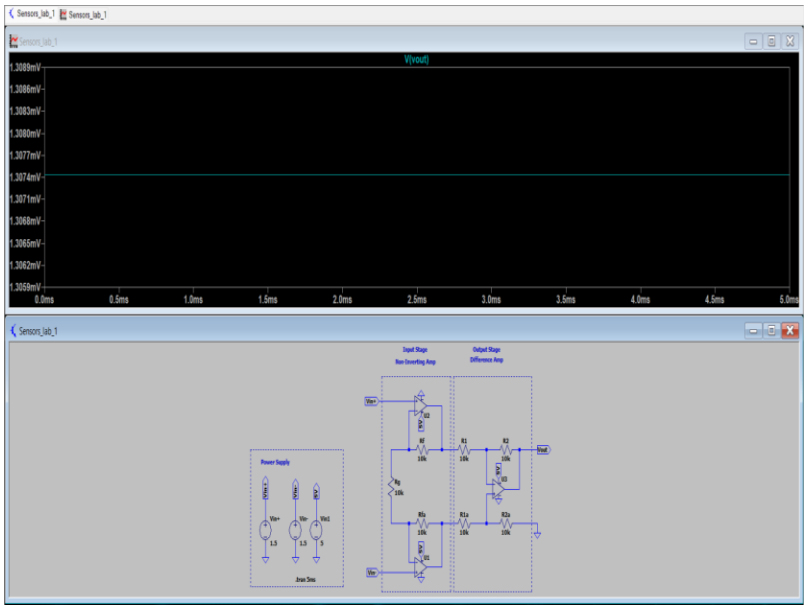


Figure 5: Simulated common-mode output, V(vout) ≈ 1.307 mV, at VIN+ = VIN− = 1.5 V.

6.1 Common-Mode Gain

$$A_c = \frac{V_{out}}{V_{cm}}$$

Measured Values:

$$V_{out} = 10.6mV$$

$$V_{cm} = 1.5$$

Calculating the common-mode gain:

$$A_c = \frac{10.6mV}{1.5}$$

$$A_c = 7.06mV$$

6.2 CMRR

$$CMRR_{dB} = 20\log_{10}\left|\frac{A_d}{A_c}\right|$$

Known Values:

$$A_c = 7.06mV$$

$$A_d = 20$$

Calculating CMRR:

$$CMRR_{dB} = 20\log_{10}\left|\frac{A_d}{A_c}\right|$$

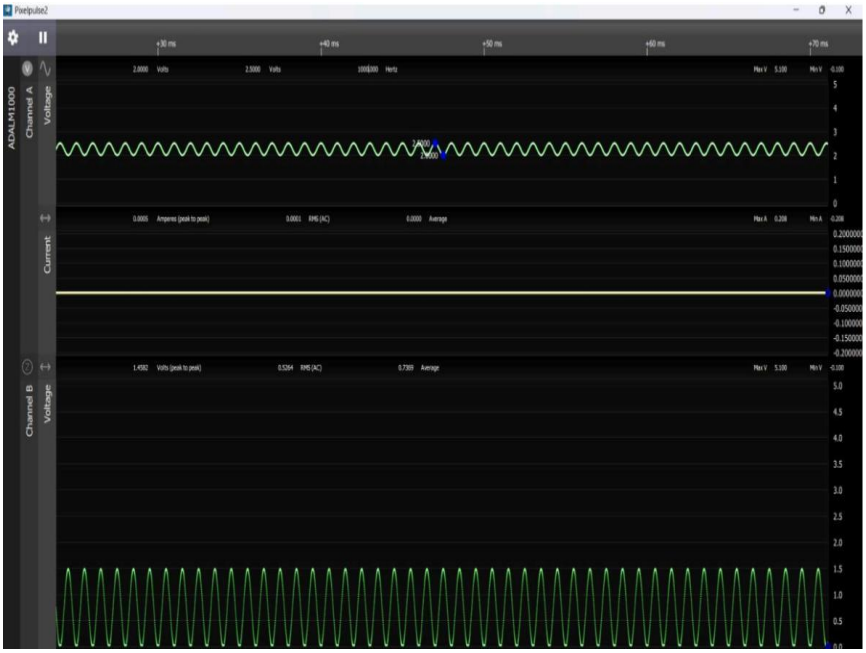
$$CMRR_{dB} = 20\log_{10}\left|\frac{20}{7.06mV}\right|$$

$$CMRR_{dB} = 20 \times 3.45222529$$

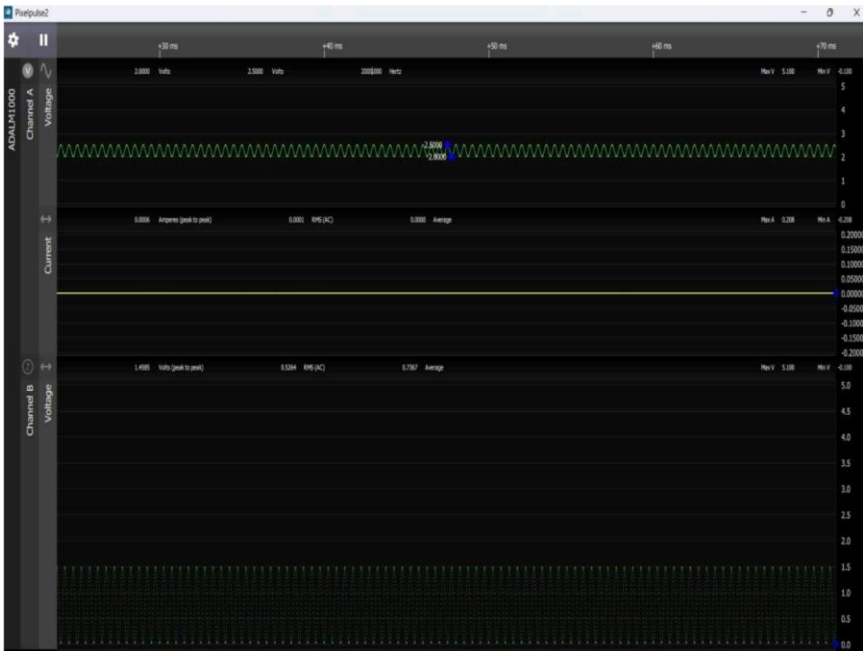
$$CMRR_{dB} = 69.0445058\ dB$$

7. Frequency Response

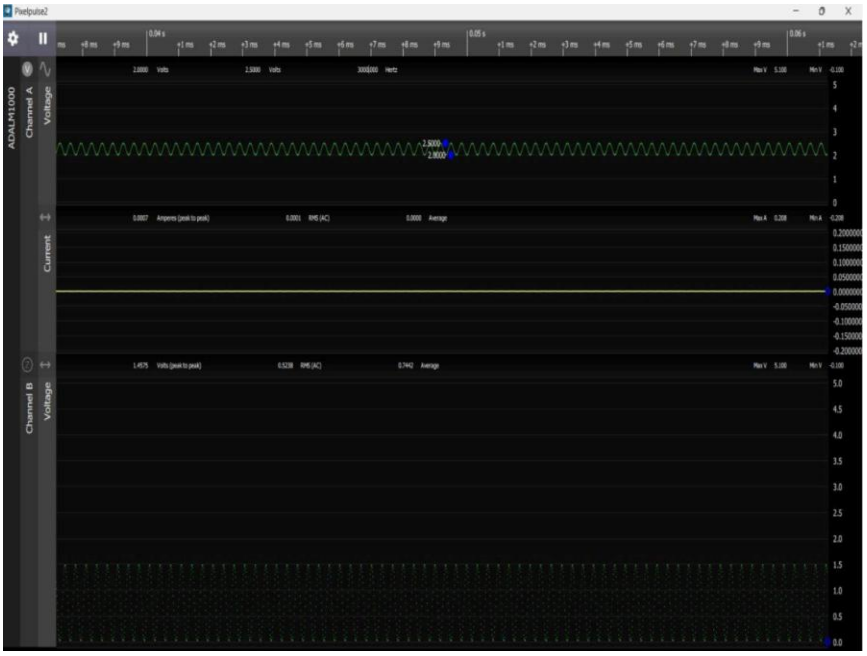
A fixed differential input $V_d = 0.5 \cdot sin(2\pi ft)\ V$ was applied and the frequency swept from 1 kHz to 5 kHz, with the resulting gain recorded at each step.



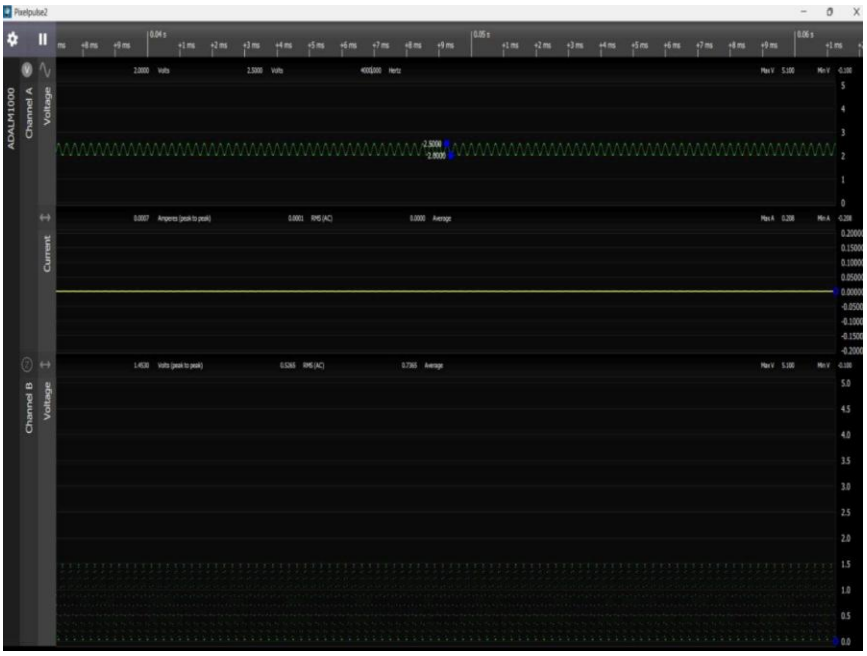
(a) f = 1000 Hz — Gain ≈ 3



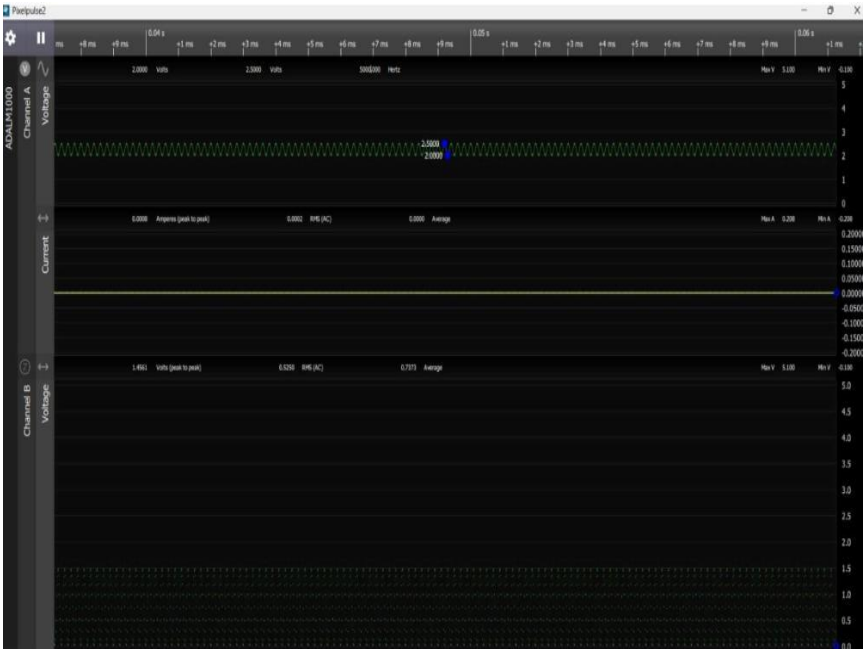
(b) f = 2000 Hz — Gain ≈ 3



(c) f = 3000 Hz — Gain ≈ 3



(d) f = 4000 Hz — Gain ≈ 3



(e) f = 5000 Hz — Gain ≈ 3

The gain remains approximately 3 across the tested 1–5 kHz range, indicating that the INA bandwidth comfortably exceeds this frequency window.

8. Differential Gain of 20

Steps 4(a) and 5 were repeated after reconfiguring the circuit for a differential gain Ad of 20, sweeping Vd from 50 mV to 200 mV in 50 mV steps.

$$A_{DG} = \left(1 + 2 * \frac{10k\Omega}{R_G}\right)$$

With $A_{DG} = 20V$ and $R = 10\ k\Omega$:

$$R_G = \frac{20k\Omega}{(20-1)}$$

$$R_G = \frac{20}{19}k\Omega$$

$$R_G = 1.05k\Omega$$

$$R_G \approx 1k\Omega\ (approximately)$$

Table 2: DC Transfer Characteristic Results (Ad = 20)

VIN+ (V)	VIN− (V)	Vd (V)	Vout,exp (V)
1.5	1.50	0.00	0.009
1.5	1.55	0.05	1.142
1.5	1.60	0.10	2.200
1.5	1.65	0.15	3.230
1.5	1.70	0.20	4.200

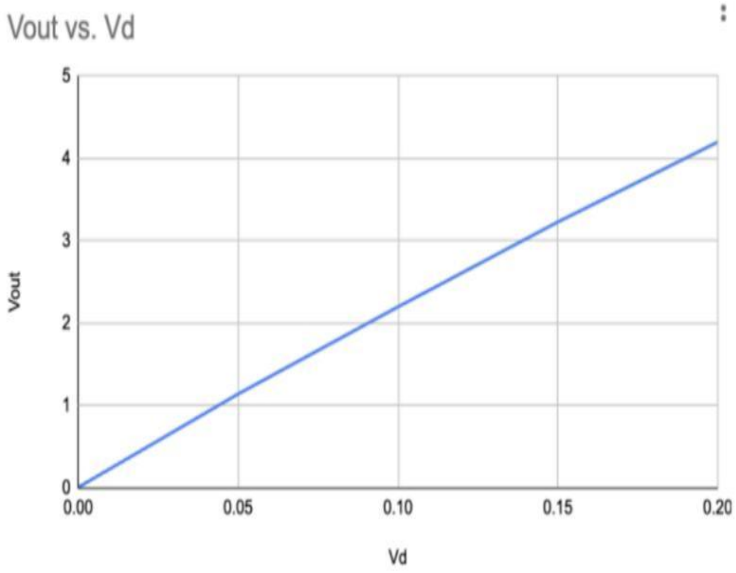


Figure 6: V_{out} vs. V_d for the $A_d = 20$ configuration — linear response consistent with the target gain.

8.1 Common-Mode Gain and CMRR (Ad = 20)

$$A_c = \frac{V_{out}}{V_{cm}}$$

$$CMRR_{dB} = 20\log_{10}\left|\frac{A_d}{A_c}\right|$$

Measured Values:

$$V_{out} = 10.6mV$$

$$V_{cm} = 1.5$$

Calculating the common-mode gain:

$$A_c = \frac{10.6mV}{1.5}$$

$$A_c = 7.06mV$$

We know $A_d = 20$:

Calculating CMRR:

$$CMRR_{dB} = 20\log_{10}\left|\frac{A_d}{A_c}\right|$$

$$CMRR_{dB} = 20\log_{10}\left|\frac{20}{7.06mV}\right|$$

$$CMRR_{dB} = 20 \times 3.45222529$$

$$CMRR_{dB} = 69.0445058\ dB$$

9. Reference-Voltage Offset (VIN+ = VIN− = 0, VREF = 2.5 V)

An expression for VOUT was derived and then verified with both simulation and hardware measurement when VIN+ = VIN− = 0 and VREF = 2.5 V.

$$V_{out} - V_{ref} = \frac{R_2}{R_1}\left(1 + \frac{2R}{R_G}\right)(V_{IN-} - V_{IN+})$$

if $V_{IN+} = V_{IN-} = 0$:

$$V_{out} = V_{ref}$$

Since VIN+ = VIN− = 0, the differential term vanishes and the amplifier simply follows the reference input:

$$V_{OUT} = V_{REF} = 2.5\,V$$

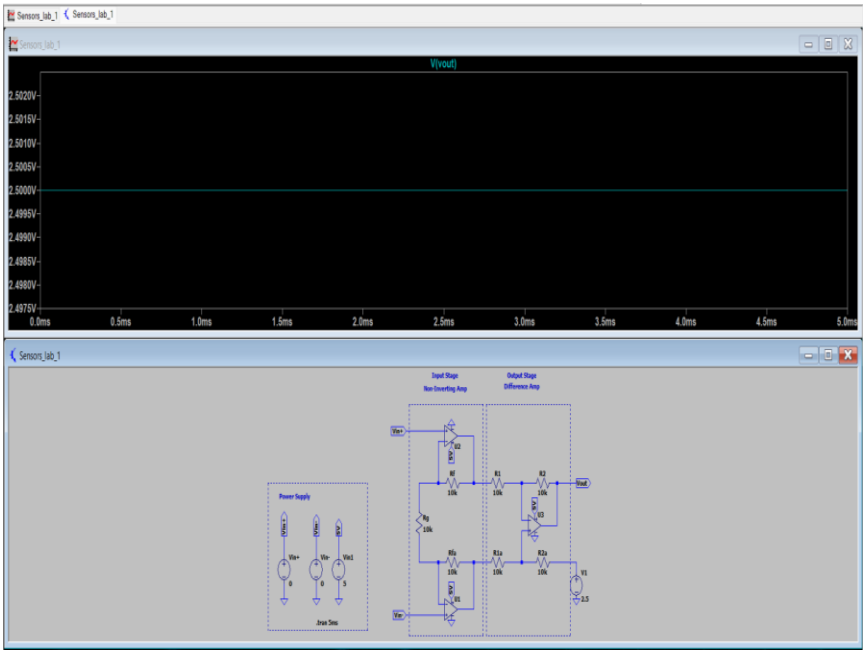


Figure 7: LTspice simulation confirming VOUT settles to 2.500 V when VIN+ = VIN− = 0 and VREF = 2.5 V.



Figure 8: Multimeter measurement of $V_{OUT} \approx 2.46\,V$, *in close agreement with the predicted 2.5 V.*